

EventSmartKafka

Adapter Configuration and Operations Guide

SAP Cloud Integration Custom Kafka Consumer Adapter

v1.0.0 RC1 - Clean Package Edition

Release profile: prod-hardening-safe with configurable Offset Commit Timeout and ESA redeploy recovery procedure

Open Source	Kafka Consumer	Cloud Connector TCP/SOCKS5	Avro JSON/XML
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1. Document Map

Section	Purpose
Installation	Import the ESA, import the demo package and prepare CPI artifacts.
Adapter Tabs	Configure connection, authentication, processing, recovery, conversion and advanced options.
Cloud Connector	Configure TCP/SOCKS5 routing and align Kafka advertised listeners.
Operations	Use the ESA redeploy recovery procedure, monitor headers and troubleshoot common errors.
Validation	Run smoke tests, stress tests and release checks before sharing the package.

2. Installation and Package Import

Use this sequence for a clean installation of the adapter and the ready-to-use iFlow package.

1. Import the EventSmartKafka ESA adapter archive into SAP Integration Suite.
2. Import the EventSmartKafka demo iFlow package only after the adapter type is available.
3. Create or update all CPI Security Material aliases used by the channels.
4. For On-Premise scenarios, configure SAP Cloud Connector TCP mappings before deploying the iFlows.
5. Open each Sender channel and replace placeholder host, port, topic, group ID, credential alias and Schema Registry values.
6. Deploy one smoke-test iFlow first. Deploy the full test matrix only after the single-channel smoke test is green.

Artifact	Required Action
ESA adapter archive	Import first. This registers the EventSmartKafka Sender adapter type in CPI.
Demo iFlow package	Import second. Replace placeholders before deployment.
Security Material	Create User Credentials, certificates or aliases required by the selected profile.
Cloud Connector mapping	Required only for Proxy Type = On-Premise. Use protocol TCP.

3. Quick Start Profiles

Scenario	Minimum Settings
Local On-Prem PLAINTEXT	Proxy Type = On-Premise; Authentication = NONE; TLS = false; Cloud Connector TCP mapping required.
On-Prem SASL_PLAINTEXT	Proxy Type = On-Premise; Authentication = SASL; TLS = false; set mechanism and credential alias.
On-Prem SASL_SSL	Proxy Type = On-Premise; Authentication = SASL; TLS = true; configure trust source and credential alias.
Cloud Kafka SASL_SSL	Proxy Type = Internet; Authentication = SASL; TLS = true; use broker host and CPI credential alias.
Avro to JSON/XML	Set conversion format, Schema Registry endpoint, schema mode and payload protection values.

4. Connection Tab

The Connection tab defines where the Kafka client starts and whether the connection is direct or routed through SAP Cloud Connector.

Field	Guidance
Bootstrap Host	Kafka bootstrap hostname. For On-Premise, use the Cloud Connector virtual host, for example sdia-kafka.
Bootstrap Port	Kafka bootstrap port. For On-Premise, use the Cloud Connector virtual port, for example 19101.
Proxy Type	Use Internet for public/cloud brokers. Use On-Premise for SAP Cloud Connector TCP/SOCKS5 routing.
Location ID	Use only when your Cloud Connector landscape requires a named location. Leave empty for the default connector.

Internet bootstrap example:
pkc-xxxxx.us-east-1.aws.confluent.cloud:9092

On-Premise virtual bootstrap example:
sdia-kafka:19101

5. Authentication Details

Authentication	TLS	Kafka Security Protocol	Credential Required
NONE	Off	PLAINTEXT	No
NONE	On	SSL	No
SASL	Off	SASL_PLAINTEXT	Yes
SASL	On	SASL_SSL	Yes

Field	Guidance
SASL Mechanism	Supported profiles include PLAIN, SCRAM-SHA-256 and SCRAM-SHA-512.
Credential Alias	CPI Security Material alias. Do not hardcode user/password values in the channel.
TLS Enabled	Enable when the broker listener requires SSL or SASL_SSL.
Trust Source	Use JVM default truststore, PEM content or CPI certificate alias according to the broker certificate chain.
Hostname Verification	Keep enabled for production-style tests. Certificate SAN must match the broker hostname seen by the Kafka client.

6. Processing Tab

Field	Guidance
Topic	Kafka topic consumed by the channel. Use domain-aligned EDCP naming where possible.
Group ID	Kafka consumer group. Changing it creates a new consumer position.
Partition	Use explicit partition for deterministic tests when needed. Leave empty only when group assignment is intended and validated.
Max Poll Records	Limit records fetched per poll. Lower values simplify recovery; higher values increase throughput.
Auto Offset Reset	EARLIEST, LATEST or NONE. Applies only when no committed offset exists for the group.
Error Handling	Stop on Error, Skip Failed Message or Retry Failed Message. Choose based on correctness vs continuity.

Policy	Behavior
Stop on Error	Failed offset is not committed. The consumer stops and avoids zombie assignment.
Skip Failed Message	Failed offset plus one is committed. Use only when poison-pill tolerance is accepted.
Retry Failed Message	Offset is committed only after successful processing. Exhausted retries fall back to stop behavior.

7. Message Recovery Tab

TIMESEEK allows controlled replay from a timestamp without manually editing broker offsets or creating a new group ID.

Mode	Use Case
Disabled	Normal consumption. No seek is applied.
Seek by Timestamp	Replay events from the first offset whose timestamp is greater than or equal to the configured timestamp.
Single Message Isolation	Read only the first matched message and restore the original committed position afterward.

Timestamp Format	Example
ISO-8601 UTC	2026-06-10T14:30:00Z
Unix epoch milliseconds	1749566200000
Unix epoch seconds	1749566200
Auto Detect	Adapter infers the input format.

8. Conversion Tab

Field	Guidance
Inbound Payload Format	NONE passes the raw payload. JSON or XML enables Avro conversion output.
Schema ID Mode	Magic Byte reads the Confluent wire-format header. Fixed Schema ID uses a configured schema ID.
Schema Registry URL	HTTPS endpoint used to fetch Avro schemas.
Schema Registry Credential Alias	CPI Security Material alias used by the Schema Registry client.
Max Payload Size	Protects the CPI worker from unexpectedly large payloads.
Schema Cache	Reduces repeated Schema Registry calls for stable schema IDs.

Payload Case	Expected Behavior
Valid Confluent wire format	Magic byte plus schema ID plus Avro payload is decoded using the resolved schema.
Wrong fixed schema ID	Adapter fails with a deterministic conversion error.
Truncated Avro payload	Adapter fails with a bounded parser error instead of silent corruption.

9. Advanced Tab

Advanced settings control network tolerance, poll behavior and offset commit behavior. These values are important under SAP Cloud Connector, CPI Trial and large-payload tests.

Field	Default / Recommendation	Purpose
Request Timeout	30s or higher for remote brokers	Kafka client request timeout.
Metadata Timeout	30s or higher for On-Premise tests	Maximum wait for broker metadata.
Offset Commit Timeout	60s for large payload and Cloud Connector tests	Maximum wait for commitSync after CPI processing.
Retry Backoff	Use conservative values in CPI Trial	Controls retry spacing for transient failures.
Poll Delay	Keep low enough to avoid zombie-looking channels	Controls ScheduledPollConsumer cadence.

10. SAP Cloud Connector TCP/SOCKS5 Setup

10.1 Cloud Connector Mapping

For On-Premise Kafka, configure Cloud to On-Premise mappings with protocol TCP. The CPI adapter uses the virtual host and port; Cloud Connector forwards to the internal host and port.

Virtual Host	Internal Host	Protocol
sdia-kafka:19101	localhost:19101	TCP
sdia-kafka:19102	localhost:19102	TCP
sdia-kafka:19103	localhost:19103	TCP

Adapter On-Premise bootstrap example:

Host = sdia-kafka

Port = 19101

Proxy Type = On-Premise

Location ID = empty unless your Cloud Connector requires it

10.2 Kafka Advertised Listener Rule

Kafka bootstrap is only the first connection. After bootstrap, the broker returns advertised listener metadata. In relay scenarios, the advertised listener must route the Kafka client back through the reachable relay path.

Local Docker validation pattern:

Cloud Connector virtual host and port: sdia-kafka:19101 to localhost:19101

Kafka external listener: 0.0.0.0:19101

Kafka advertised listener: localhost:19101

Adapter bootstrap: sdia-kafka:19101 with Proxy Type = On-Premise

10.3 Local Relay Lifecycle Note

If repeated deploy and redeploy cycles are performed during development, a local relay port may appear locked. This is usually visible as Address already in use. Restarting SAP Cloud Connector may not release it because the local port is held on the CPI/JVM side.

- This is not a Kafka broker overload symptom.
- This is not an Avro or Schema Registry symptom.
- This is a CPI/OSGi local relay lifecycle troubleshooting case observed with custom TCP relay adapters.

10.4 Mandatory Channel Redeploy After ESA Update

During validation, a reliable operational recovery was observed: after deploying a new ESA/custom adapter version, redeploy every iFlow/channel that uses the adapter before changing Cloud Connector ports.

1. Deploy the new EventSmartKafka ESA.
2. Redeploy all iFlows and channels using EventSmartKafka.
3. Wait for the channels to restart and recreate their endpoints.
4. Test the same relay ports again.
5. Change ports only if Address already in use persists after the full redeploy cycle.

Why it works: redeploying dependent channels forces CPI/OSGi to recreate adapter endpoints and consumers. In testing, this broke stale local relay state and avoided unnecessary Cloud Connector port changes.

11. Runtime Headers and Monitoring

The adapter injects diagnostic headers into the Camel Exchange. These headers are available to downstream iFlow steps and are useful in logs, Content Modifier, trace and custom troubleshooting.

Header	Meaning
x-sdiakafka-topic	Kafka topic consumed.
x-sdiakafka-partition	Kafka partition number.
x-sdiakafka-offset	Kafka offset delivered to CPI.
x-sdiakafka-key	Kafka message key when present.
x-sdiakafka-timestamp	Kafka record timestamp in epoch milliseconds.
x-sdiakafka-conversionFormat	Active conversion mode such as NONE, JSON or XML.
x-sdiakafka-resolved-schema-id	Schema ID used by the Avro decoder when applicable.
x-sdiakafka-error-code	Structured adapter error code when an error path is triggered.

12. Troubleshooting Matrix

Symptom	Most Likely Cause	Action
Address already in use	Local CPI relay port is still held by an old adapter/channel runtime.	Redeploy all iFlows using the adapter. Change port only if the error persists.
SOCKS5 read timed out	Cloud Connector path or SOCKS5 handshake did not respond within timeout.	Keep as retryable. Verify Cloud Connector reachability and mapping.
Kafka metadata timeout	Broker returned advertised listener metadata that CPI cannot reach.	Fix advertised.listeners and Cloud Connector virtual mapping.
Commit timeout after processing	Record was processed, but commitSync exceeded Offset Commit Timeout.	Increase Offset Commit Timeout. Start with 60s; use 120s for heavy 2 MB stress tests if needed.
Schema Registry gzip/parse issue	Registry response compression or invalid schema response.	Verify Schema Registry URL, credentials, timeout and gzip handling.
Wrong fixed schema ID	Configured schema ID does not match the Confluent wire-format schema ID.	Use Magic Byte mode or correct the Fixed Schema ID.

13. Example Channel Profiles

Profile	Example Settings
On-Prem PLAINTEXT	Host sdiakafka, Port 19101, Proxy On-Premise, Authentication NONE, TLS false.
On-Prem SASL_SSL SCRAM-512	Host sdiakafka, Port 19103, Proxy On-Premise, Authentication SASL, TLS true, Mechanism SCRAM-SHA-512, credential alias configured.
Avro to JSON 2 MB	Conversion JSON, Schema Registry enabled, Magic Byte mode, Max Payload Size above expected payload, Offset Commit Timeout 60s or higher.

14. Validation Sequence

1. Compile and package the adapter ESA.
2. Import the ESA into SAP Integration Suite.
3. Redeploy all iFlows that use EventSmartKafka after any ESA update.
4. Run a single-message smoke test.
5. Run a 10-message smoke test.
6. Run conservative 100k validation.
7. Run large stress validation such as 800k messages after cooldown.
8. Check CPI Monitor for completed messages, failed messages and retries.
9. Capture screenshots and terminal output for release evidence.

15. SDIA / EDCP / ODCP Naming Guidance

Use a consistent semantic grammar across Kafka topics, CPI packages, iFlows and security aliases. This is not mandatory for the adapter to run, but it improves governance and troubleshooting.

Artifact	Example
EDCP topic	acme.sales.otc.order.created
ODCP package	acme.br.sales.otc.i
iFlow name	id00.otc.kafka.ordercreated.cpi.none.plaintext.op
Security alias	cr.sales.otc.basic.kafka

16. Release Checklist

- ESA imported successfully.
- All dependent iFlows redeployed after ESA update.
- Security Material aliases exist and match channel configuration.
- Cloud Connector TCP mappings are reachable.
- Kafka advertised listeners align with the relay path.
- Offset Commit Timeout configured for large payload tests.
- Avro conversion tested with expected payload sizes.
- CPI Monitor evidence captured for smoke and stress runs.
- Known limitations documented as demo/technical preview, not SAP-certified production SLA.

17. Final Notes

EventSmartKafka v1.0.0 RC1 is provided as an open-source technical demo and public review baseline. It is not an SAP-certified product and does not provide an enterprise SLA. Production adoption requires validation by the adopting organization.

18. CPI Integration Configuration for Custom Headers

EventSmartKafka injects diagnostic headers into the Camel Exchange using the x-sdiakafka prefix. These headers are available inside the iFlow, but CPI monitoring visibility depends on the iFlow runtime/integration configuration. If the headers are not allowed or configured, they may not appear in the monitor or may not be available for search and troubleshooting.

Configure the iFlow runtime/integration settings before deploying the channel. In SAP Integration Suite, open the iFlow configuration area used by your tenant for header propagation or custom header visibility, then allow the EventSmartKafka diagnostic headers.

Configuration Item	Recommended Value
Header prefix	x-sdiakafka-
Wildcard when supported	x-sdiakafka-*
Do not rely only on dot notation	The RC1 headers use hyphenated names such as x-sdiakafka-topic, not x-sdiakafka.topic.
Fallback when wildcard is not accepted	Enter the explicit header list shown below.
When to configure	Before deploying the iFlow/channel that uses the adapter.
Why it matters	Without this configuration, the adapter can still process records, but the diagnostic headers may not be visible in CPI monitoring or downstream troubleshooting steps.

18.1 Minimal Recommended Header Set

For most demo, troubleshooting and validation scenarios, start with this minimal set. It keeps the monitor useful without creating excessive noise.

Header	Purpose	Recommended for Monitor
x-sdiakafka-topic	Kafka topic consumed.	Yes
x-sdiakafka-partition	Kafka partition number.	Yes
x-sdiakafka-offset	Kafka offset delivered to CPI.	Yes
x-sdiakafka-key	Kafka record key when present.	Yes
x-sdiakafka-timestamp	Kafka record timestamp in epoch milliseconds.	Yes
x-sdiakafka-conversionFormat	Active conversion mode such as NONE, JSON or XML.	Yes
x-sdiakafka-resolved-schema-id	Schema ID used by the Avro decoder when applicable.	Yes for Avro
x-sdiakafka-error-code	Structured adapter error code when an error is raised.	Yes for errors

18.2 Full Diagnostic Header Set

For full technical validation, allow the complete EventSmartKafka header set. This is useful during RC validation, stress tests, troubleshooting and SAP Community demo evidence collection.

Header	Meaning
x-sdiakafka-topic	Kafka topic consumed.
x-sdiakafka-partition	Kafka partition number.
x-sdiakafka-offset	Kafka offset delivered to CPI.
x-sdiakafka-key	Kafka record key when present.
x-sdiakafka-timestamp	Kafka record timestamp in epoch milliseconds.
x-sdiakafka-conversionFormat	Active conversion mode such as NONE, JSON or XML.
x-sdiakafka-schema-mode	Schema resolution mode such as Magic Byte or Fixed Schema ID.
x-sdiakafka-resolved-schema-id	Schema ID resolved for Avro conversion.

x-sdiakafka-error-code	Structured adapter error code.
x-sdiakafka-error-message	Short adapter error message.
x-sdiakafka-error-detail	Detailed technical diagnostic message.
x-sdiakafka-error-handling	Active error handling policy.
x-sdiakafka-stop-on-error	Whether stop-on-error behavior was triggered.
x-sdiakafka-recovery-mode	Configured recovery mode.
x-sdiakafka-recovery-read-mode	Recovery read behavior.
x-sdiakafka-seek-applied	Whether TIMESEEK was applied.
x-sdiakafka-seek-input	Original timestamp input configured for recovery.
x-sdiakafka-seek-timestamp-ms	Parsed recovery timestamp in epoch milliseconds.
x-sdiakafka-seek-offset	Offset resolved by Kafka offsetsForTimes when applicable.

18.3 Configuration Procedure

1. Open the integration flow that uses the EventSmartKafka Sender channel.
2. Go to the iFlow runtime or integration configuration area used by your tenant for allowed headers, propagated headers or custom header visibility.
3. Add x-sdiakafka-* if wildcard patterns are accepted by the screen.
4. If the wildcard is not accepted, add the explicit header names from the full diagnostic list.
5. Deploy the iFlow after changing the header configuration.
6. Produce one message and verify that the headers are visible in the message monitor or trace.
7. For the demo package, repeat the configuration for every iFlow that must expose adapter diagnostics.

18.4 Monitoring Guidance

Use topic, partition and offset as the primary operational coordinates. Together, they identify the exact Kafka record handled by CPI. For Avro scenarios, also expose conversion format and resolved schema ID. For error scenarios, expose the structured error code and error message.

Do not expose sensitive values such as credential passwords, bearer tokens, Schema Registry secrets or payload content. Credential aliases may be useful for internal troubleshooting but should not be enabled as public monitoring fields unless the organization accepts that visibility.

18.5 Release Checklist Addition

- Integration Configuration allows x-sdiakafka-* or the explicit x-sdiakafka header list.
- A single-message smoke test confirms that topic, partition and offset headers are visible.
- Avro channels confirm conversionFormat and resolved-schema-id visibility when applicable.
- Error scenario tests confirm x-sdiakafka-error-code and x-sdiakafka-error-message visibility.